

2.1 The power of surveys

We observe that the aforementioned challenges in studying LM opinions also arise when attempting to measure human opinions for research or policymaking. The primary approach for the latter currently is to use *public opinion surveys*. According to Pew Research: “Much of what the country [US] knows about its media usage, labor and job markets, educational performance, crime victimization, and social conditions is based on data collected through polls.” These surveys address the first of the three challenges with the help of experts, who identify topics of public interest and carefully design questions to capture the nuances of the topic. To tackle the difficulties associated with analyzing open-ended responses, survey designers craft the questions to be *multiple-choice*. Finally, surveys determine humans’ opinions on these topics through extensive polling of the public at large. (A further discussion of the meticulous data collection process followed by survey designers is provided in Appendix A.1.) These factors make public opinion surveys an ideal testbed to study LM opinions, and our work develops methods for querying LMs with these surveys, as well as evaluation metrics for quantifying their alignment w.r.t. human opinions.

2.2 Our framework

We now put forth a general methodology to convert multiple-choice public opinion surveys into datasets for evaluating LM opinions. Consider a survey with a set of questions Q , where a question q has a set of possible answers $A(q)$. Each question is also categorized into a set of topics (it can have multiple associated topics), such that the questions belonging to a topic T (e.g., “guns” for Figure 1) are denoted by Q_T . As part of the survey, each question is presented to a carefully chosen pool of participants, where every individual (h) must select one answer $F(h, q)$.

To use this data for our study, we need to obtain the *human opinion distribution* against which we can compare LMs. For a question, we can build this distribution by aggregating the responses over a set of human respondents H , i.e., $D_H(q) = \sum_{h \in H} w_h F(h, q)$. During aggregation, we can weight respondents uniformly $w_h = 1/|H|$, or if available, using weights assigned by the survey to correct sampling biases ($\sum_{h \in H} w_h = 1$). In this work, we will consider two different sets of respondents – all survey respondents (O) or a demographic group such as “Democrats” (G). We use $D_O(q)$ and $D_G(q)$ to denote the associated marginal opinion distributions respectively.

2.3 Instantiating OpinionQA

We now apply this methodology to the annual “*American Trends Panel*” (ATP) polls conducted by Pew research to build the OpinionQA dataset (details in Appendix A.2). Concretely, we use 15 ATP polls, chosen to cover a range of topics such as privacy, political views, and health. Each poll contains two key objects that we will use for our analysis: a set of multiple-choice questions (typically ~ 100) and answers from respondents (typically on the order of thousands) from across the US along with their demographic information (Appendix Table 1). We use individual survey responses – in conjunction with demographic information and participant weights – to obtain the per-question overall $D_O(q)$ and group-level $D_G(q)$ human opinion distributions for each of 60 demographic groups (Appendix Table 2). Pew surveys often touch upon a broad range of (often overlapping) issues—both ATP-W26 and ATP-W92 have questions about guns. Thus, we further aggregate the dataset questions into the 23 coarse and 40 fine-grained topic categories shown in Appendix Table 3.

Note: While our methodology is general, the OpinionQA dataset itself is English and US-centric. Thus, our subsequent analysis is limited to the US populace and demographic groups within (see

Section 6 for a discussion).

3 Measuring human-LM alignment

We now discuss how to probe language model opinions on questions from our OpinionQA dataset and compare them to the previously-obtained human opinion distributions.

3.1 Interfacing with models

Prompting the model. Due to the multiple-choice nature of samples in our dataset, we can use standard prompting approaches used for traditional question answering (QA) tasks (Hendrycks et al., 2020; Liang et al., 2022). Concretely, we format each question into the prompt template shown in Figure 1. Unless otherwise specified, we present the options in the order they are provided by the survey designers, which captures the ordinal structure of the options – e.g., “A great deal” to “Not at all” in Figure 1. We then evaluate LMs on these questions in two settings, distinguished by the additional context provided to the model.

When evaluating representativeness (Sections 1 and 4.1), the goal is to understand the LM’s *default* opinion distribution, and we prompt the model using this standard QA template without any added context. In contrast, measuring steerability (Section 4.2) involves testing the model’s ability to adapt to a particular demographic group. In this *steered* setting, we thus prepend additional context to the prompt describing the group that we want the model to emulate. We consider three approaches to supply demographic information to the LM (examples in Figure 1):

1. QA: The group information is provided as a response to a previous multiple-choice survey question, using the phrasing used by Pew to collect this information.
2. BIO: The group information is provided as a free-text response to a biographic question (e.g., asking about party affiliation), akin to Argyle et al. (2022).
3. PORTRAY: The LM is instructed to pretend to be a member of said group (e.g. pretend you are a Democrat), similar to the crowd-sourcing design of Kambhatla et al. (2022).

Extracting the output distribution. In contrast to factual QA tasks, there is no “correct” answer in our setting. Instead, for a model m , we are interested in the *distribution of model opinions* $D_m(q)$ for each question across its corresponding set of answer choices. To obtain this distribution, we prompt the model and obtain the next-token log probabilities. Specifically, we measure the log probabilities assigned to each of the answer choices (e.g., ‘A’, ‘B’, ... in Figure 1) – ignoring all other possible completions (See Appendix A.3 for details).

For reasons that we will discuss in Section 3.2, we treat the refusal and non-refusal answer choices (“E” and “A”-“D” in Figure 1) separately. Concretely, to compute $D_m(q)$, we exponentiate and normalize the scores for all answer choices *except refusal*. Then, for questions with a refusal option, we also measure the model’s refusal probability as the ratio of the exponentiated log probability of refusal vs. the exponentiated cumulative log probabilities for all the choices (e.g., $e^{lp(E)} / \sum_{o \in \{A,B,C,D,E\}} e^{lp(o)}$ for the Figure 1 example).

3.2 Evaluating the model’s response

Aggregating human responses from the opinion surveys, as well as probing LMs, provide us with a set of opinion distributions $D(q)$ (i.e., overall, group-level and per-LM) over the answer choices.

To answer our question of whose opinions LMs reflect, we must now define a similarity measure over pairs of such distributions. Although we could use any distributional divergence to compare two distributions, there are some subtleties in the structure of survey questions that we would like to capture. Specifically, unlike standard QA benchmarks, the answer choices to survey questions typically have an ordinal structure (e.g., ranging from “A great deal” to “Not at all”, along with a refusal option in Figure 1). This means that divergences for non-metric probability measures such as the Kullback-Liebler or total variation can provide misleading estimates of disagreement. For instance, if all humans answered “A great deal”, a model that assigns all its probability mass to “A fair amount” and another one that assigns all its mass to “Not at all” would be incorrectly deemed equally similar based on such measures.

We thus choose the 1-Wasserstein distance ($\mathcal{WD}$), which for a pair of distributions D_1 and D_2 , is defined as the minimum cost for transforming D_1 into D_2 . Note that here the cost of transformation accounts for the similarity between answer choices. To project the ordinal answer choices to a metric space suitable for $\mathcal{WD}$, we simply map them to the corresponding positive integers (e.g., {‘A’: 1, ‘B’: 2, ..., ‘D’: 4} for Figure 1). There are two exceptions: (i) due to its non-ordinal nature, we omit the ‘Refused’ option (if present) in computing $\mathcal{WD}$ and compare human and model refusals separately, and (ii) if the last option is hedging (e.g., “Neither” and “About the same”), we map it to the mean of the remaining ordinal keys (see Appendix A.4 for details).

Measuring opinion alignment. We define *alignment* between two opinion distributions D_1 and D_2 on a set of questions Q as:

$$\mathcal{A}(D_1, D_2; Q) = \frac{1}{|Q|} \sum_{q \in Q} 1 - \frac{\mathcal{WD}(D_1(q), D_2(q))}{N - 1} \quad (1)$$

Where, N is the number of answer choices (excluding refusal) and the normalization factor $N - 1$ is the maximum $\mathcal{WD}$ between any pair of distributions in this metric space. This metric is bounded between 0 and 1, with a value of 1 implying a perfect match between the two opinion distributions. In our study, we use this metric to compare the LM opinion distribution D_m to that of all survey respondents (D_O) and that of specific groups (D_G).

On the use of the term *alignment*. We use the term *alignment* to describe our metric as it measures one aspect of alignment — alignment of opinions and preferences between LMs and humans. Crucially, in contrast to prior work, our work treats human alignment as an inherently subjective quantity that depends on who it is measured against, rather than it being a single quantity that can be improved. In fact, based on our definition, higher human-LM alignment to certain groups might not always be desirable (e.g., matching racist views) or even possible (e.g., aligning with both Democrats and Republicans on abortion) – see Section 6.

4 Whose views do current LMs express?

We now evaluate existing models on OpinionQA and analyze their opinion agreement with respect to people in the US. We study a set of 9 LMs—with different providers (OpenAI and AI21 Labs), scales (350M to 178B parameters), data collection, and training strategies. These models can be roughly grouped into (i) base LMs, that have only been pre-trained on internet data (ada, davinci, davinci, j1-grande and j1-jumbo), and (ii) human feedback (HF)-tuned LMs that have